

Understanding User Sentiment in Recipe Feedback

Venus Zayoud

About the Dataset

- Source: [UCI Machine Learning Repository](#)
- Contains over 18,000 recipe reviews and user feedback records.
- Includes:
 - Review text
 - Star ratings
 - User reputation
 - Engagement metrics (thumbs up, thumbs down, replies)
 - Recipe information
 - Timestamps

Provides both textual data and structured metadata, making it suitable for sentiment analysis and machine learning experiments.

Why This Project?

- I chose this project because it combines **Natural Language Processing (NLP)** and **Machine Learning** with real user-generated data.
- Online reviews contain valuable information about **user opinions, satisfaction, and engagement**.
- The project provides an opportunity to explore the complete NLP workflow, from text preprocessing and sentiment analysis to classification and evaluation.
- Understanding review sentiment can help businesses and content platforms gain insights into user preferences and improve decision-making.

Data Preparation & Exploration

- **Data Preparation**

- Data cleaning
- Handling missing values
- Feature engineering
- Timestamp processing

- **Exploratory Data Analysis**

- Review length distribution
- Rating distribution
- User engagement patterns
- Correlation analysis

NLP Pipeline

- **Text Processing**
 - Tokenization
 - Stopword removal
 - Lemmatization
- **Sentiment Analysis**
 - VADER sentiment scoring
 - Sentiment label generation
 - Comparison with star ratings
- **Vectorization**
 - CountVectorizer
 - TF-IDF

Modeling & Evaluation

- **Classification Models**

- Logistic Regression
- Multinomial Naive Bayes
- Linear SVC

- **Handling Imbalance**

- SMOTE oversampling

- **Evaluation**

- Accuracy
- Weighted F1 Score
- Classification Reports
- Confusion Matrices

Why These Models?

- **Logistic Regression:** strong and interpretable baseline for text classification.
- **Naive Bayes:** simple, fast, and widely used in NLP applications.
- **Linear SVC:** effective for high-dimensional sparse text features.
- **SMOTE:** balances minority classes and improves model fairness.

Results & Impact

- VADER sentiment scores generally align with user star ratings.
- The dataset is highly imbalanced toward positive reviews.
- Logistic Regression and Naive Bayes achieved reasonable accuracy, but struggled with minority classes.
- Linear SVC + SMOTE achieved the best overall performance:
 - Accuracy $\approx 75\%$
 - Weighted F1 ≈ 0.78
- Review sentiment can help platforms identify dissatisfied users and improve recommendations.

Challenges

- Strong class imbalance (majority of reviews are 5-star).
- Short and informal reviews make sentiment detection difficult.
- Minority classes remain harder to classify accurately.
- High-dimensional text features increase computational requirements.

Conclusions & Future Work

Conclusions & Future Work

- Recipe reviews contain meaningful sentiment information.
- Classical NLP and machine learning can effectively classify review sentiment.
- Class imbalance has a major impact on performance.

Future Work

- Explore [transformer](#) models (BERT, DistilBERT).
- Test alternative imbalance-handling techniques.
- Build recommendation systems using text embeddings.

References

- [UCI Machine Learning Repository – Recipe Reviews and User Feedback Dataset](#)
- [Natural Language Processing with Classification and Vector Spaces](#) (Coursera)
- [Machine Learning for Beginners](#) (Microsoft)
- [NLTK Documentation](#)
- [Scikit-Learn Documentation](#)
- [Natural Language Processing with Python](#) (Bird, Klein & Loper)
- ReDI Connect mentoring materials provided by Dr. Raji Ghawi